

Experiment list. The table lists all experiments of the paper. *Establishes* states the claim the experiment supports. Status: **done** = analyzed; **data in hand** = collected, analysis pending re-framing; **running** = currently executing; **stale, re-run** = old data from a superseded configuration, to be re-collected; **to run** = not started.

Experiment	Establishes	Status
E0 Retrieval / replay micro-benchmark	Cost anatomy: full step 72.5 ms vs. hit ≈ 14 ms (Stage-1 10.3 + retrieval 4.15); origin of the IR floor 0.152	done
E1 Main frontier: threshold sweeps, 2 benchmarks $\times$ 2 teachers	The deployable lower bound on $R(\varepsilon)$; 50% @ <1 pp / 63% @ <5 pp (Spatial), 30% @ ~ 1 pp (LIBERO-10); difficulty dependence of R	done LIBERO running RoboCasa
E2 Oracle arm: retrieval by simulator state	Second term of the sandwich; separates coverage from key loss	to run (highest leverage)
E3 Library-size scaling, 6 tiers $\times$ 2 suites	R grows with the library (0.27 $\rightarrow$ 0.52 on LIBERO-10); pure replay stays below the teacher; no saturation claim	done
E4 Temporal structure of hits	Stickiness 0.91–0.96 vs. marginals 0.28–0.70; runs of 9–12 steps; 93–98% single-trajectory lockstep	done
E5 Score discriminability (AUROC, offline)	The thresholded score is informative for outcome; descriptive only	to run (offline, cheap)
E6 Cross-scene transfer (RoboCasa365, held-out kitchens)	Varies $\mathcal{M}$ with $(\pi_T, \mathcal{L}, \varphi)$ fixed; preliminary: robust on large-motion tasks	running
E7 Key-space ablations: E7a source (<i>main text</i> : direct evidence for the title claim) / E7b depth, E7c fusion (appendix)	The teacher’s own representation is an adequate index (the φ argument)	to run (offline replay)
E8 Warm start, three-way control	Experience-guided step reduction vs. blind reduction at equal skipped budget	stale, re-run
E9 Small-model comparison (ACT / SmolVLA in the hit slot)	Regime map: distillation vs. replay; acquisition-cost accounting	data in hand
E10 Trained-router comparison	Whether a learned scheduler raises the lower bound over the threshold rule within $\mathcal{C}(\mathcal{I}_\varphi)$	running
E11 Serving mini-bench: $N \in \{4, 8, 16\}$ concurrent episodes per GPU	Measured anchor for the robots-per-GPU projection (1/IR)	to run
E12 History-augmented negatives	Depth-1 keys suffice; history terms do not help (one table in the appendix)	data in hand
E13 Real-robot validation (SO-101): full lifecycle, 3 tasks by motion scale, ≈ 330 episodes	The mechanism functions under real perception and physics; measured hit/miss latency; removes the simulation-only limitation	to run (teacher fine-tuning first)

The shaded rows (E1, E2) are plotted in a single figure: the oracle arm overlays the main frontier as a second curve per panel. New collection required: E2, E11, the E8 refresh, and E13. E5 and E7 are offline computations on logged data. E1 (RoboCasa cells), E6, and E10 are in progress.